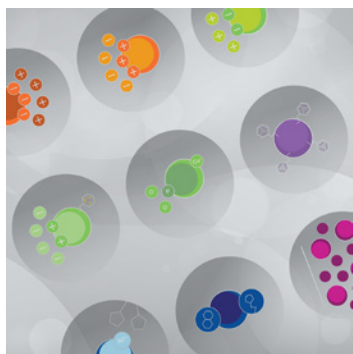




CHT™ Ceramic Hydroxyapatite XT Media



CHT CERAMIC HYDROXYAPATITE

- High physical and chemical stability
- Unique selectivity
- Efficient single-step clearance of aggregates and other impurities
- Straightforward column packing
- Full regulatory support

Robust, Long-Lasting Media for Repeated Use and Exceptional Resolution

Introduction

CHT Ceramic Hydroxyapatite Media is a spherical, macroporous form of hydroxyapatite that has been sintered at high temperatures to yield a physically and chemically robust support. It is ideal for large-scale bioprocess applications, providing consistent results at scales of at least 1.8 meters. CHT Ceramic Hydroxyapatite provides unique separation properties along with unmatched selectivity and resolution. It can be used in applications such as purification of enzymes, monoclonal and polyclonal antibodies of various classes, antibody fragments, and bispecific antibodies; efficient isolation and purification of viruses and virus-like particles (VLPs); and separation of supercoiled DNA from linear duplexes as well as single stranded from double-stranded DNA.

High Stability and Unique Selectivity

The technical features of CHT Ceramic Hydroxyapatite XT Media (CHT XT) are listed in Table 1. It is designed to provide added robustness and lifetime. The chemical composition of CHT XT is $(\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2)$ and it is a mixed-mode media. Biomolecules can interact with CHT through calcium affinity interactions and/or cation exchange interactions. Refer to bulletin 6086, CHT Applications Guide (bio-rad.com/CHTGuide) for details on both these interactions.

Table 1. Properties of CHT Ceramic Hydroxyapatite XT.

Property	Description
Function	Mixed-mode: cation exchange (phosphate) and affinity (calcium)
Functional group	Ca^{2+} , PO_4 , OH
Mean particle size	40
Dynamic binding capacity	17–25 mg Lysozyme/g* ≥ 60 mg/ml mAb G** (pI 9.2)
Recommended linear flow rate	50–300 cm/hr
Maximum operating pressure	10 bar (150 psi)
Tap settled density	0.67 g/ml
Compression factor	Incompressible
pH stability	6.5–14
Shipping form	Dry powder
Regeneration	400 mM sodium phosphate, pH ≥7.5 0.4–1 M potassium phosphate, pH ≥7.5 1 M trisodium phosphate, pH 11–12
Sanitization	1 N NaOH
Autoclavability (bulk)	121°C, 20 min in phosphate buffered saline, pH 7
Storage conditions	0.1 N NaOH
Chemical compatibility	Stable in all commonly used organic solvents Stable in all commonly used buffer components, except for chelating agents (citrate, EDTA, etc.) Stable in solutions at pH 6.5 or greater
Shelf life	5 years

* At 500 cm/hr with 10 mM sodium phosphate, pH 6.8.

** At 100 cm/hr with 5 mM sodium phosphate, 25 mM sodium chloride, pH 7 in a 0.5 x 9.5 cm column.

A small amount (up to 5 mM) of sodium phosphate should be added to all unbuffered solutions as a counter ion.

Under appropriate buffer conditions, CHT XT Media can be used for a large number of purification cycles (Table 2). Chromatographically, CHT XT Media maintains the unique separation properties expected from CHT Media, including efficient resolution (Figure 1) and aggregate removal (Figure 2). Structurally, the three CHT Ceramic Hydroxyapatite types differ in their pore diameters and, therefore, in protein separation uses. This offers flexibility in selecting the CHT type that provides maximum benefit in a given separation. CHT Type I and XT have higher protein binding capacities than CHT Type II due to their large surface areas. CHT Type II, however, provides higher capacity for large molecules such as IgM and viruses as it has the largest pore diameter of the three CHT types.

Long Lifetime Ensures Repeated Reusability

CHT XT Media can be used for over 70 cycles under the conditions listed in Table 2, making it ideal for manufacturing processes.

Table 2. Column lifecycle study of CHT XT. A 20 x 20 cm column was packed with CHT XT and cycled continuously at 140 cm/hr using buffer protocols I or II until a column backpressure of 3 bar was reached.

Protocol	Number of Cycles
Protocol I	139
Protocol II	74

Protocol I:

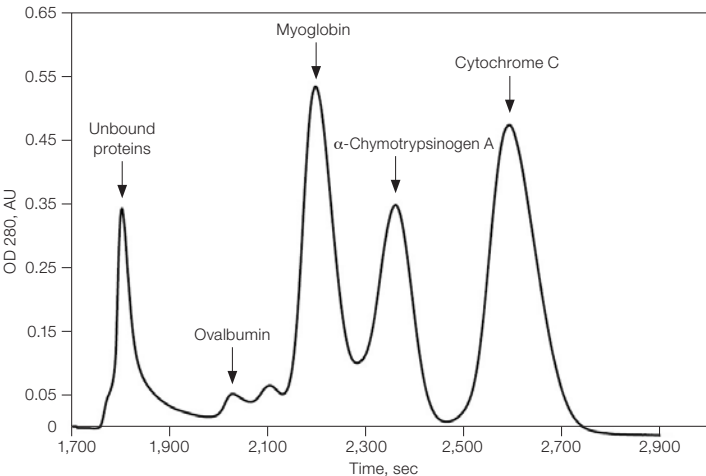
Pre-equilibration: 0.4 M sodium phosphate, pH 7.0, 4 column volumes (CV)
 Equilibration, loading, and wash: 5 mM sodium phosphate, 0.1 M sodium chloride, pH 6.5, 15 CV
 Elution: 5 mM sodium phosphate, 0.55 M sodium chloride, pH 6.5, 4 CV
 Stripping: 0.4 M sodium phosphate, pH 7, 3 CV
 Sanitization: 1 M sodium hydroxide, 3 CV

Protocol II:

Pre-equilibration: 0.4 M sodium phosphate, pH 6.5, 4 CV
 Equilibration, loading, and wash: 5 mM sodium phosphate, pH 6.5, 15 CV
 Elution: 5 mM sodium phosphate, 0.55 M sodium chloride, pH 6.5, 4 CV
 Stripping: 0.4 M sodium phosphate, pH 6.5, 3 CV
 Sanitization: 1 M sodium hydroxide, 1 M sodium chloride, 3 CV

Efficient Resolution of Proteins

A protein mixture was used to demonstrate the resolution properties of CHT XT. Figure 1 shows the efficient separation of the mixture, which consisted of four different proteins.



Flow rate: 1.57 ml/min
Pre-equilibrium: 400 mM sodium phosphate, pH 6.8, 4 CV
Loading buffer: 5 mM sodium phosphate, pH 6.8, 15 CV
Post-load wash: 5 mM sodium phosphate, pH 6.8, 1 CV
Elution: Linear gradient elution 0–75% 400 mM sodium phosphate, pH 6.8, over 15 CV
Stripping: 400 mM sodium phosphate, pH 6.8, 3 CV
Sanitization: 1 M sodium hydroxide, 3 CV

Fig. 1. Separation of protein standards. A mixture of the following protein samples in 10 ml of 5 mM sodium phosphate, pH 6.8, was used for purification: 120 mg of ovalbumin, 75 mg of myoglobin, 60 mg of α-chymotrypsinogen A, 75 mg of cytochrome C. One hundred microliters of this mixture was loaded onto a 0.7 x 5.4 cm column with a packed bed volume of 2 ml of CHT XT.

Aggregate Removal during Monoclonal Antibody Purification

A monoclonal antibody sample (mAb G) was used to test the aggregate removal and antibody purification capability of CHT XT. The mAb G sample contained a significant amount (~28%) of aggregate as shown by size exclusion chromatography (SEC) analysis (Figure 2A). This sample was purified using CHT XT (Figure 2B) and the eluate from the resultant peak was again analyzed by SEC (Figure 2C). Complete separation of the mAb aggregates from the monomers was observed, demonstrating that CHT XT is highly effective at aggregate removal during monoclonal antibody purification.

For an additional study of aggregate removal from another mAb (mAb S), see [bulletin 7109](#), Removing the Aggregates of an Acidic Monoclonal Antibody with CHT Ceramic Hydroxyapatite XT Media.

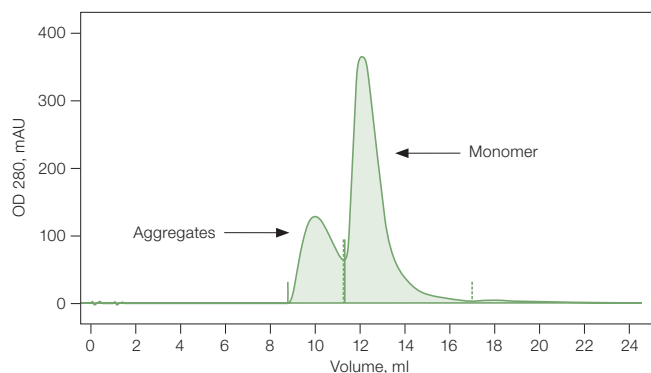


Fig. 2A. Size exclusion chromatography (SEC) of mAb G load. SEC showing the level of mAb G aggregates in the load. mAb G was loaded on a 10 x 300 mm Bio-Rad ENrich SEC 650 Column with a packed bed volume of 23.56 ml equilibrated in PBS. OD 280 (—).

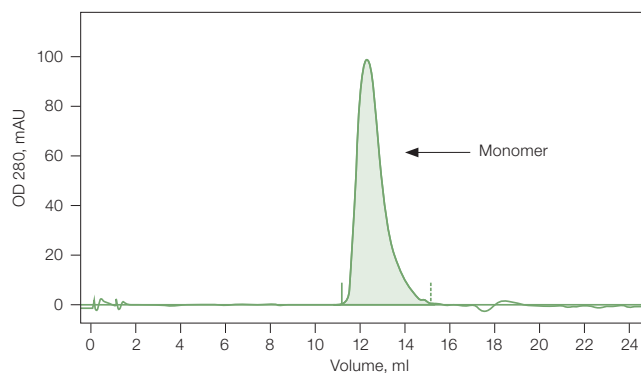
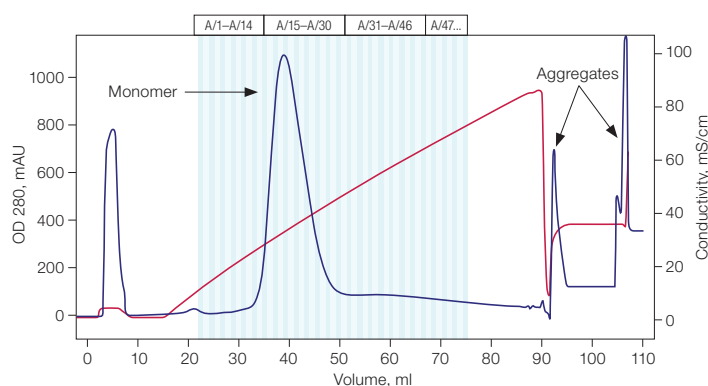


Fig. 2C. Final SEC profile of mAb G monomer pool. The SEC profile of the pooled fractions from CV 13–27 (A/13–A/27 in Figure 2B) confirms aggregate clearance from the mAb G monomer. OD 280 (—).



Sample load: 40 mg mAb G sample with aggregates
Column: 1.85 ml CHT XT packed in a 0.5 x 9.4 cm column
Flow rate: 1 ml/min
Equilibration: 10 mM sodium phosphate, pH 6.5, 10 CV
Wash: 10 mM sodium phosphate, pH 6.5, 3 CV
Elution: Linear gradient elution 0–100% 10 mM sodium phosphate, 1 M sodium chloride, pH 6.5
Fraction collection: 1 ml fraction size over 40 CV
Stripping: 0.4 M sodium phosphate, pH 7, 4 CV
Sanitization: 1 M sodium hydroxide, 3 CV

Fig. 2B. mAb G purification profile. Elution profile showing separation of the monomer from higher molecular weight impurities. OD 280 (—); conductivity (—).

Viral Clearance

CHT XT viral clearance was evaluated using xenotropic murine leukemia virus (XMuLV), an enveloped mRNA retrovirus. Studies were conducted with a third party in accordance with ICH Q5A(R2)* guidelines. Purified IgG monoclonal antibody (mAb) was prepared in 5 mM NaPO₄, 20 mM NaCl, pH 6.5 to a concentration of 5 mg/ml. The samples were spiked with virus stock solution and loaded at >25 mg protein/ml of resin onto a prepacked 5 ml Foresight™ CHT XT Column (Bio-Rad, catalog #12003149). The mAb was eluted using ten column volumes (CV) at 300 cm/hr with a 0–100% linear gradient of 5 mM NaPO₄, 1 M NaCl, pH 6.5. The log₁₀ reduction factor was >4.86 logs.

* International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use guidance entitled Q5A(R2) Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin.

Storage

Columns packed with CHT XT Media can be stored in 0.1 M sodium hydroxide. In dry powder form, CHT XT should be stored in a closed container at room temperature. When sealed in the original container, unused CHT XT Media can be stored in the dry form for at least five years at room temperature.

Ordering Information

Catalog # Description

Bulk Resins

CHT Ceramic Hydroxyapatite XT Media

12002457	CHT Ceramic Hydroxyapatite XT, 40 µm, 10 g
12002454	CHT Ceramic Hydroxyapatite XT, 40 µm, 100 g
12002456	CHT Ceramic Hydroxyapatite XT, 40 µm, 1 kg
12002455	CHT Ceramic Hydroxyapatite XT, 40 µm, 5 kg

Prepacked Screening Tools

Foresight Prepacked Chromatography Filter Plates*

12003151	Foresight CHT XT Plates, 40 µm, 20 µl
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Foresight RoboColumn Units**

12003152	Foresight CHT XT RoboColumn Units, 40 µm, 200 µl
12003148	Foresight CHT XT RoboColumn Units, 40 µm, 600 µl

Foresight Prepacked Columns

12003150	Foresight CHT XT Column, 40 µm, 1 ml
12003149	Foresight CHT XT Column, 40 µm, 5 ml

EconoFit Columns

12009261	EconoFit CHT XT Column, 1 x 1 ml, 7 x 25 mm
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Econofit Kits

12009315	EconoFit Aggregate Clearance Kit, includes 1 ml each of Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, CHT XT Media
12009314	EconoFit Large Biomolecule Kit, includes 1 ml each of Nuvia Q; Nuvia HP-Q; Nuvia cPrime; Nuvia aPrime 4A; CHT II, 40 µm; and CHT XT Media
12009313	EconoFit mAb Kit, includes 1 ml each of Nuvia S, Nuvia Q, Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, and CHT XT Media

Process Prepacked Columns

Foresight Pro Columns

12014725	Foresight Pro Column, CHT XT, 40 µm, 0.2 L, 5 cm ID x 10 cm length
12014726	Foresight Pro Column, CHT XT, 40 µm, 0.4 L, 5 cm ID x 20 cm length
12014727	Foresight Pro Column, CHT XT, 40 µm, 0.5 L, 8 cm ID x 10 cm length
12014728	Foresight Pro Column, CHT XT, 40 µm, 1.0 L, 8 cm ID x 20 cm length
12014729	Foresight Pro Column, CHT XT, 40 µm, 0.8 L, 10 cm ID x 10 cm length
12014730	Foresight Pro Column, CHT XT, 40 µm, 1.5 L, 10 cm ID x 20 cm length
12014731	Foresight Pro Column, CHT XT, 40 µm, 1.3 L, 13 cm ID x 10 cm length
12014732	Foresight Pro Column, CHT XT, 40 µm, 2.7 L, 13 cm ID x 20 cm length
12014733	Foresight Pro Column, CHT XT, 40 µm, 3.1 L, 20 cm ID x 10 cm length
12014734	Foresight Pro Column, CHT XT, 40 µm, 6.3 L, 20 cm ID x 20 cm length
12014735	Foresight Pro Column, CHT XT, 40 µm, 4.5 L, 24 cm ID x 10 cm length
12014736	Foresight Pro Column, CHT XT, 40 µm, 9.0 L, 24 cm ID x 20 cm length
12014737	Foresight Pro Column, CHT XT, 40 µm, 8.5 L, 33 cm ID x 10 cm length
12014738	Foresight Pro Column, CHT XT, 40 µm, 17 L, 33 cm ID x 20 cm length

* Package size: 2 x 96-well plates.
 ** Package size: one row of eight columns.

Technical Assistance

A regulatory support file is available upon request. Bio-Rad Laboratories Inc. is an ISO 13485 registered corporation.

For detailed information on process development, refer to bulletin 6086, CHT Applications Guide ([bio-rad.com/CHTGuide](https://www.bio-rad.com/CHTGuide)).

Visit [bio-rad.com/CHTXT](https://www.bio-rad.com/CHTXT) for more information and to request samples.

Visit us at [bio-rad.com/ProcessResins](https://www.bio-rad.com/ProcessResins) for more information on Bio-Rad’s complete line of process chromatography supports.

For additional information and technical assistance, contact your local Bio-Rad office or email our process specialists at process@bio-rad.com. In the U.S. and Canada, call 1-800-4BIORAD.

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